incredibly complex situation. As Alice insisted in Chapter Eight, the worlds aren't fundamental. Rather, they're *emergent*.

Emergence in this sense does not refer to events unfolding over time, as when a baby bird emerges from its egg. It's a way of describing the world that isn't completely comprehensive, but divides up reality into more manageable chunks. Notions like rooms and floors are nowhere to be found in the fundamental laws of physics—they're emergent. They are ways of effectively describing what's going on even if we lack perfect knowledge of each and every atom and molecule around us. To say that something is emergent is to say that it's part of an approximate description of reality that is valid at a certain (usually macroscopic) level, and is to be contrasted with "fundamental" things, which are part of an exact description at the microscopic level.

In the Laplace's demon thought experiment, we imagine a vast intelligence that would know all the laws of physics and the exact state of the world, as well as having unlimited computational capacity. To the demon, everything that is, was, and ever will be is completely known. But none of us is Laplace's demon. In reality, we have at best partial information about the state of the world, and quite limited computational capacity. None of us looks at a cup of coffee and sees every particle in every atom; we see some coarse macroscopic features of the liquid and the cup. But that can be all the information we need to have a useful discussion about the coffee, and to predict its behavior in a variety of circumstances. A cup of coffee is an emergent phenomenon.

The same thing can be said for worlds in Everettian quantum mechanics. For a quantum version of Laplace's demon, with exact knowledge of the quantum state of the universe, there would never be any need to divide the wave function into a set of branches describing a collection of worlds. But it is enormously convenient and helpful to do so, and we're allowed to take advantage of this convenience because the individual worlds don't interact with one another.

That doesn't mean that the worlds aren't "real." Fundamental versus emergent is one distinction, and real versus not-real is a completely separate one. Chairs and tables and cups of coffee are indubitably real, as they describe true patterns in the universe, ones that organize the world in ways that reflect the underlying reality. The same goes for Everettian worlds. We choose to invoke them when carving up the wave function for our